

Lecture 10.2: Minimization of DFA

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Hopcroft's Algorithm

Add dead state and transitions to it if necessary.

(Now, every state has an outgoing edge on every symbol.)

Π = initial partitioning

loop

$\Pi_{\text{NEW}} = \text{Refine}(\Pi)$

if ($\Pi_{\text{NEW}} = \Pi$) then break

$\Pi = \Pi_{\text{NEW}}$

endLoop

Construct DFA_{MIN}

- Each group in Π becomes a state
- Choose one state in each group (throw all other states away)
- Preserve the edges out of the chosen state
- Deal with start state and final states
- If desired...
 - Remove dead state
 - Remove any state unreachable from the start state

Let's apply this algorithm on an example,

Initial Partitioning:

$\Pi_1 = (A B C D) (E)$

Consider (A B C D)

Consider "a"

Break apart? **No**

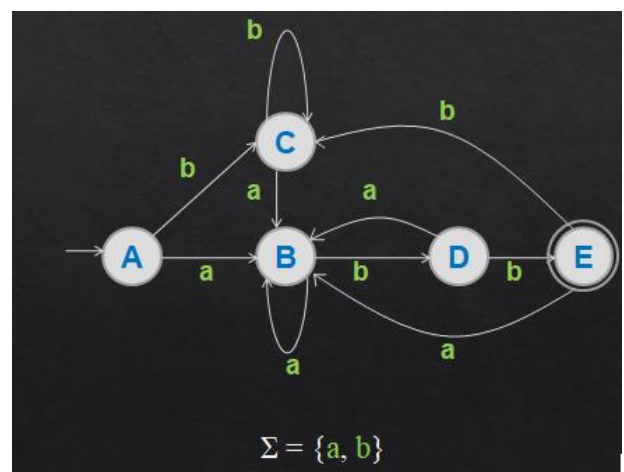
Consider "b"

Break apart? **Yes**

(A B C) (D)

Consider (E)

Not possible to break apart.



New Partitioning:

$$\Pi_2 = (A B C) (D) (E)$$

Consider (A B C)

Consider "a"

Break apart? No

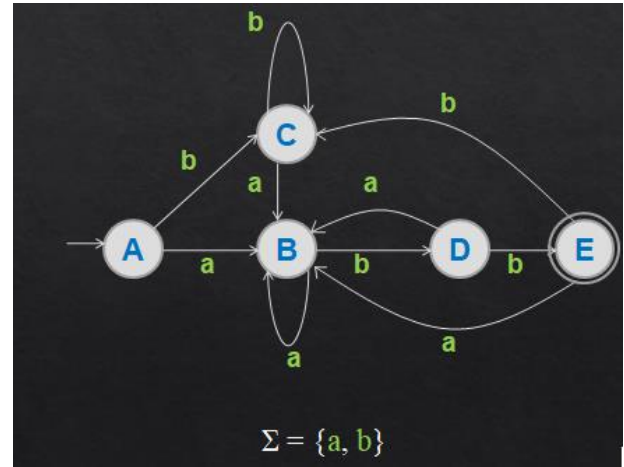
Consider "b"

Break apart? Yes

$$(A C) (B)$$

Consider (D)

Not possible to break apart.



Another New Partitioning:

$$\Pi_3 = (A C) (B) (D) (E)$$

Consider (A C)

Consider "a"

Break apart? No

Consider "b"

Break apart? No

Consider (B)

Not possible to break apart.

So, $\Pi_3 = (A C) (B) (D) (E)$ is the final Partitioning.

The Final Minimal DFA:

